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Question 1: Define Entity and Entity Set with Examples from daily life:

Ans: Entity: An entity is a real-world object or thing for which data is collected. In ER Model entity is represented by a rectangle.

Example: Student, Vehicle

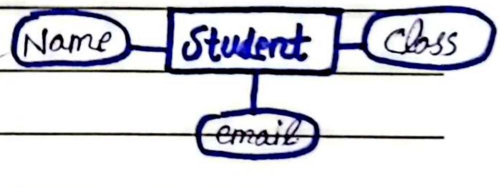
Entity Set: An Entity set is a collection of similar types of entities.

Examples: → All students in a class → Student entity set.
→ All cars in a showroom → car entity set.

Question 2: What is an Attribute? Explain types.

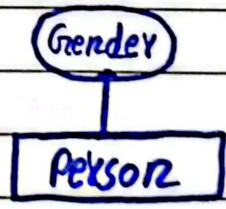
Answer: The characteristics of an entity are called Attributes or Properties. Attributes are represented as oval.

Example: Attribute of student Entity are Name, class, email etc.

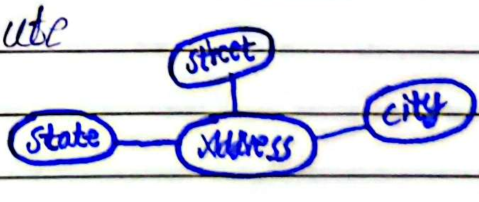


Types of Attributes:

→ **Simple Attribute:** An attribute that cannot be subdivided into smaller components. **Example:** A person can have only one gender.



→ **Composite Attribute:** An attribute that can be sub-divided into smaller components.



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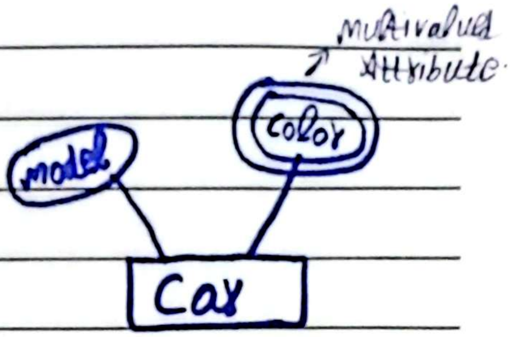
→ **Multivalued Attribute:**

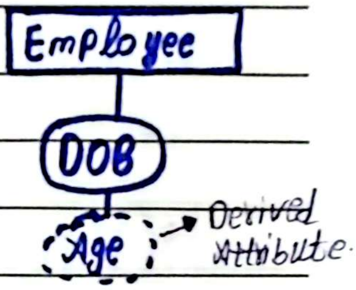
An attribute that contain two or more values. It is represented by double-line-oval.



Example: Employe may have more than one skills.

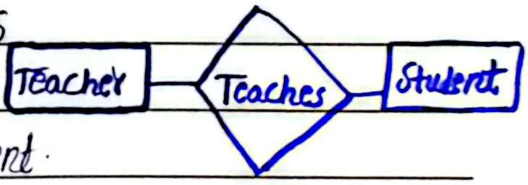
→ **Derived Attribute:**

An Attribute that is not stored in database but derived from another value. **e.g:** Age is derived from Date of Birth.



Question 3: Define Relationship and Relationship Set?

Ans: Logical connection between different Entities are called Relationship. It is represented using Diamond Symbol.



Example: A teacher teaches the student.

→ **Relationship Set:**

A relationship set is a collection of similar relationships.

Example: All teaching relationships b/w teachers and subjects.

Question 4: Explain the difference b/w strong entity and Weak entity:

Ans: **Weak entity:** An entity that can exist only if another entity exists is known as weak entity. It is represented by double-line-square.

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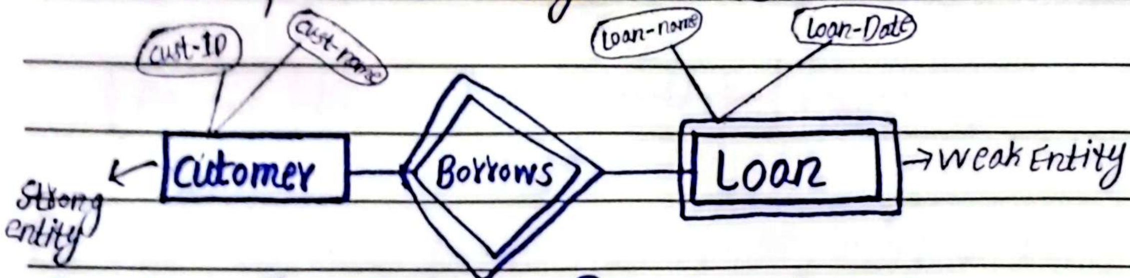
Example: Loan (Depends on Employee).

→ Strong Entity:

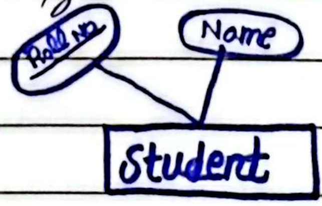
An entity that can exist without depending upon the existence of another entity. It is also called as Parent.



Question 5: What is Primary key? Why is it important in an ER model?

Ans: A Primary key is an attribute (or set of attributes) that uniquely identify each entity in an entity set.

Example: Student → Roll Number



→ Importance:

- Uniquely Identifies each record.
- It Prevents duplicate data.
- Helps in creating relationships between entities.
- It ensures data Integrity.

Question 6: Define Cardinality Constraints (1:1, 1:M, M:N) with examples:

Ans: The cardinality constraint specifies the number of instances of one entity that can be associated with each instance of the other entity.

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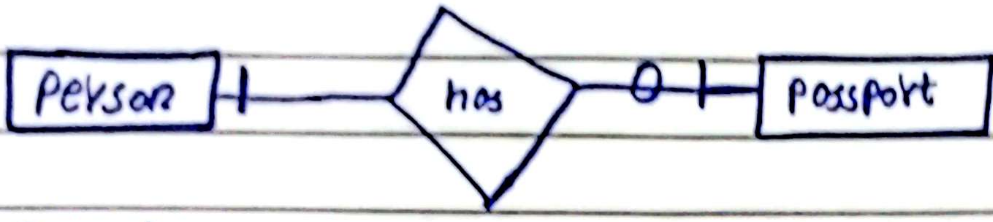
Types:

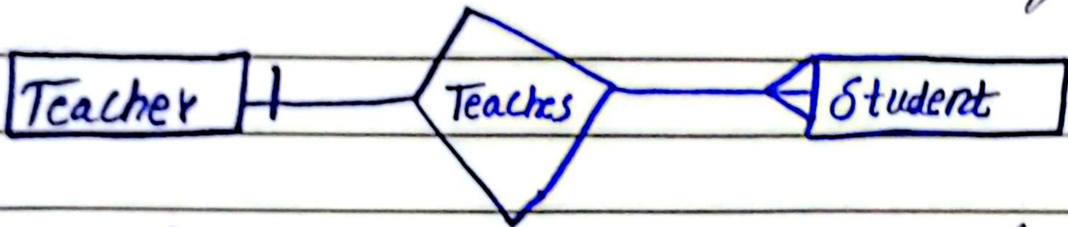
1. One to one:

One entity is related to only one entity.

Example: One Person has one Passport



2: One to Many: One entity is related to many entities **Example:** One teacher teaches many students



3- Many to Many: Many entities are related to many entities **e.g:** Students enroll in many courses and courses have many students.

Question 7: What is Participation Constraint? Explain total and Partial Participation:

Ans: Participation Contraint defines whether All entities must Participate in a relationship or not.

Types:

→ **Total Participation:**

All entities must Participate in the relationship.

Example: Every student must be enrolled in a course

2- Partial Participation:

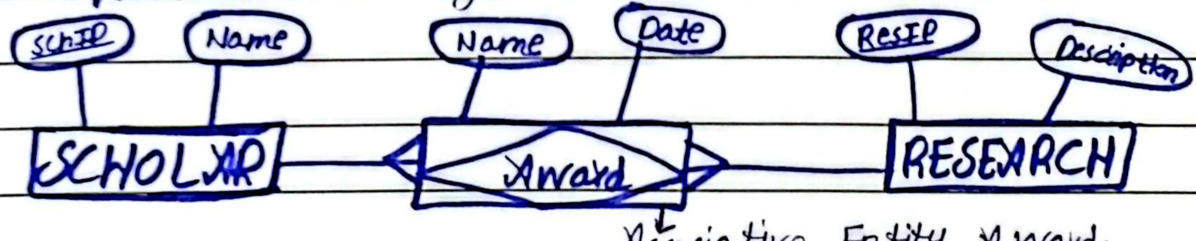
Some entities may Participate and some may not Participate in the relationship.

Example: Some employees may not be assigned to any Project.

Question 8: Explain Associative (Bridge) entity with example:

Answer: Associative entity is a type of entity that associates the instances of one entity or many entity types with one anothers. It act as a bridge b/w entities.

Example: A scholar gets an award on his research.



Associative Entity Award.

Question No 9: Differentite b/w Entity, attribute and relationship:

Ans: Entity: An entity is a real-world object or thing for which data is collected. An Er model entity is represented by a rectangle.

Example: Student, Vehicle.

→ Attribute: The characteristics of an entity are called Attributes or Properties. Attributes are represented as Oval.

Example: Attribute of student Entity are Name, class, email etc.

→ **Relationship:**

The logical connection b/w entities are called Relationship. **Example:** A teacher teaches the student.

Question 10: Convert a simple ER model into relational Schema.

Answer: ER Model:

- **Student** (Student-ID, Name).
- **Course** (Course-ID, Course-Name)
- **Relationship**: Enrolls (Many to Many)

→ **Relational Schema:**

1- **Student Table:** Student (Student-ID, Name).
2- **Course Table:** Course (Course-ID, Course-Name)
3- **Enrollment Table:** Enrollment (Student-ID, Course-ID).

→ **Explanation:**

- Each entity becomes a table.
- Primary keys remain the same.
- Many to many relationship is converted into a separate table.

Question no 11: Identify entities, attributes, and relationships in a Hospital System.

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→ **Entities:** • Patient • Doctor • Nurse
• Room

→ **Attributes:**
• Patient → Patient-ID, Name, Age, Disease.
• Doctor → Doctor-ID, Name, Specialization.
• Nurse → Nurse-ID, Name
• Room → Room-No, Type.

→ **Relationships:**
• Patient is treated by Doctor.
• Nurse assists Doctor.
• Patient is admitted in Room.

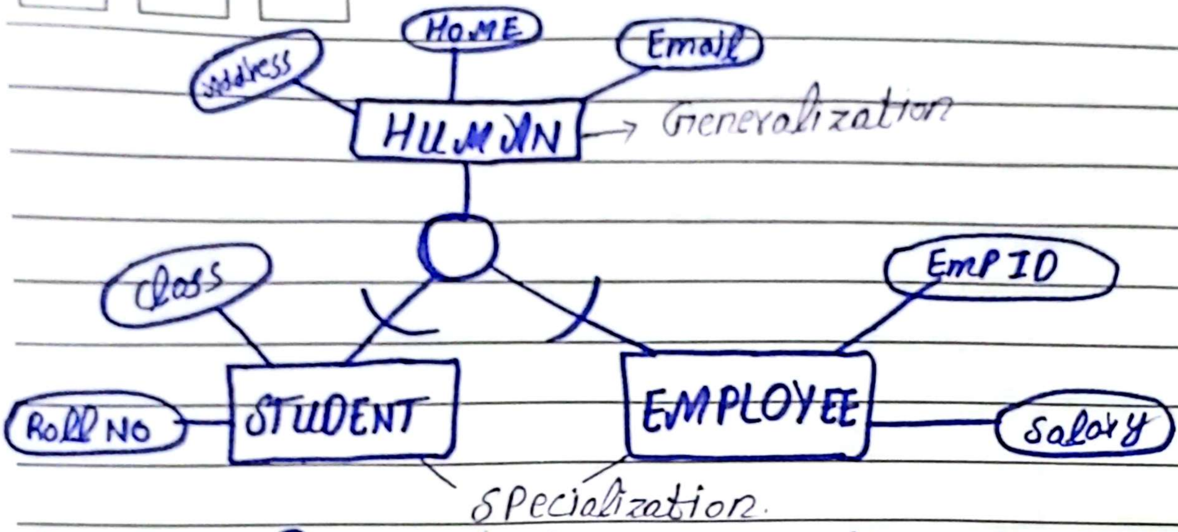
Q No 12: Explain the concept of specialization and generalization:

Answer: Specialization:

Specialization is a process of identifying more specific entity types. **For example** STUDENT and LAWYER are more specific entity types than HUMAN.

→ **Generalization:**

Generalization is a process of identifying more general entity type. **For example:** HUMAN is more general entity type than STUDENT or [redacted] LAWYER. The entity type HUMAN contains attributes that are more general than other two entity Type



Question 13: What is Aggregation in ERModel? Explain with Example:

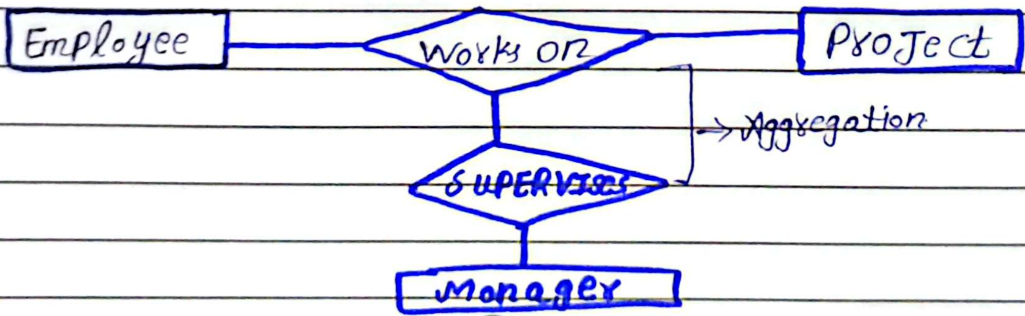
Answer: Aggregation:

Aggregation is a Process where a relationship between entities is treated as a single entity to relate with another entity.

OR: Aggregation is a process in which a relationship is treated as an entity so that it can participate in another relationship.

Example: Employee work on Project → (this is a relationship).

- Manager Supervises that work Here "works on" is treated like an entity. The Manager connects to it.



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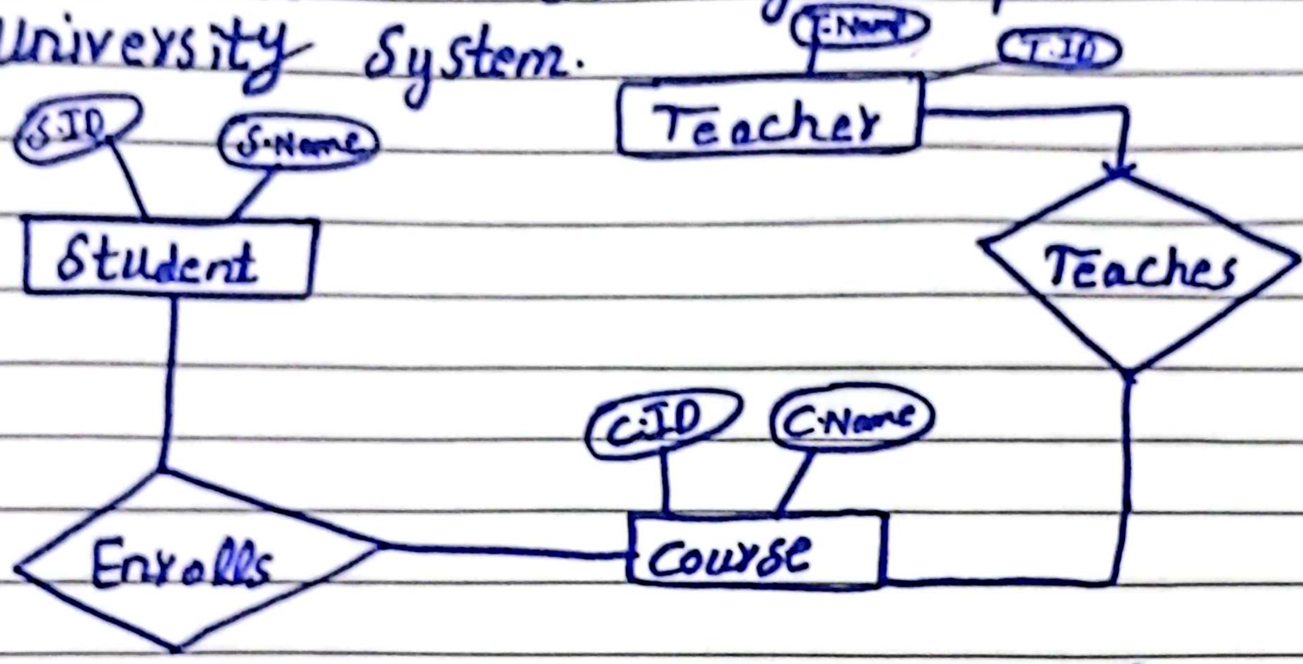
Question 8: Draw an ER diagram for a University System.



Question 9: Draw an ER diagram for a Library Management System.